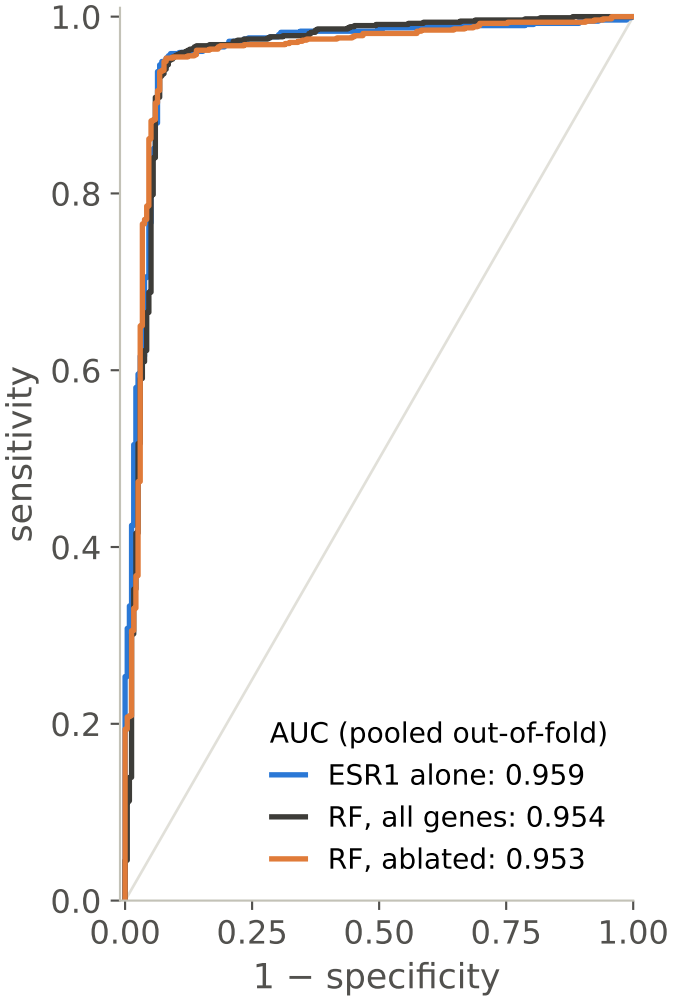
Models match but do not beat the receptor gene alone (ESR1 0.96, ERBB2 0.85).

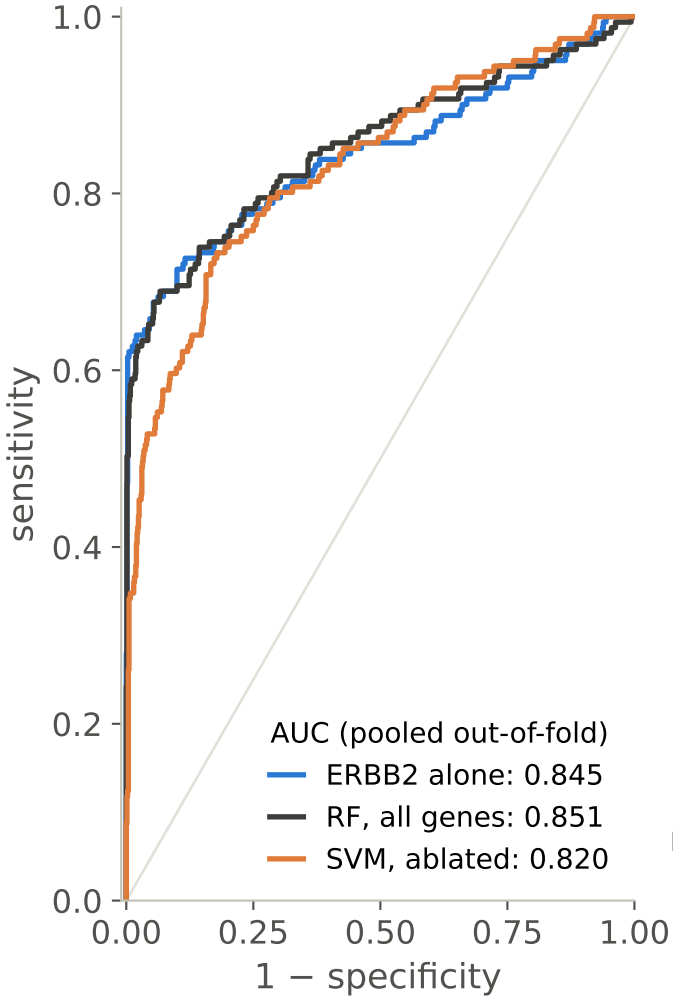
Without receptor genes, 17q12 amplicon and PAM50: ER unchanged (0.95), HER2 down to 0.78–0.82.

Best model vs gene alone, Nadeau-Bengio corrected t on 15 paired folds: ER RF ΔAUC -0.004 (p 0.69); HER2 RF ΔAUC +0.008 (p 0.66).
Shuffled labels: MCC -0.01 (ER), +0.04 (HER2). Filtering, scaling and feature selection refitted inside every training fold.

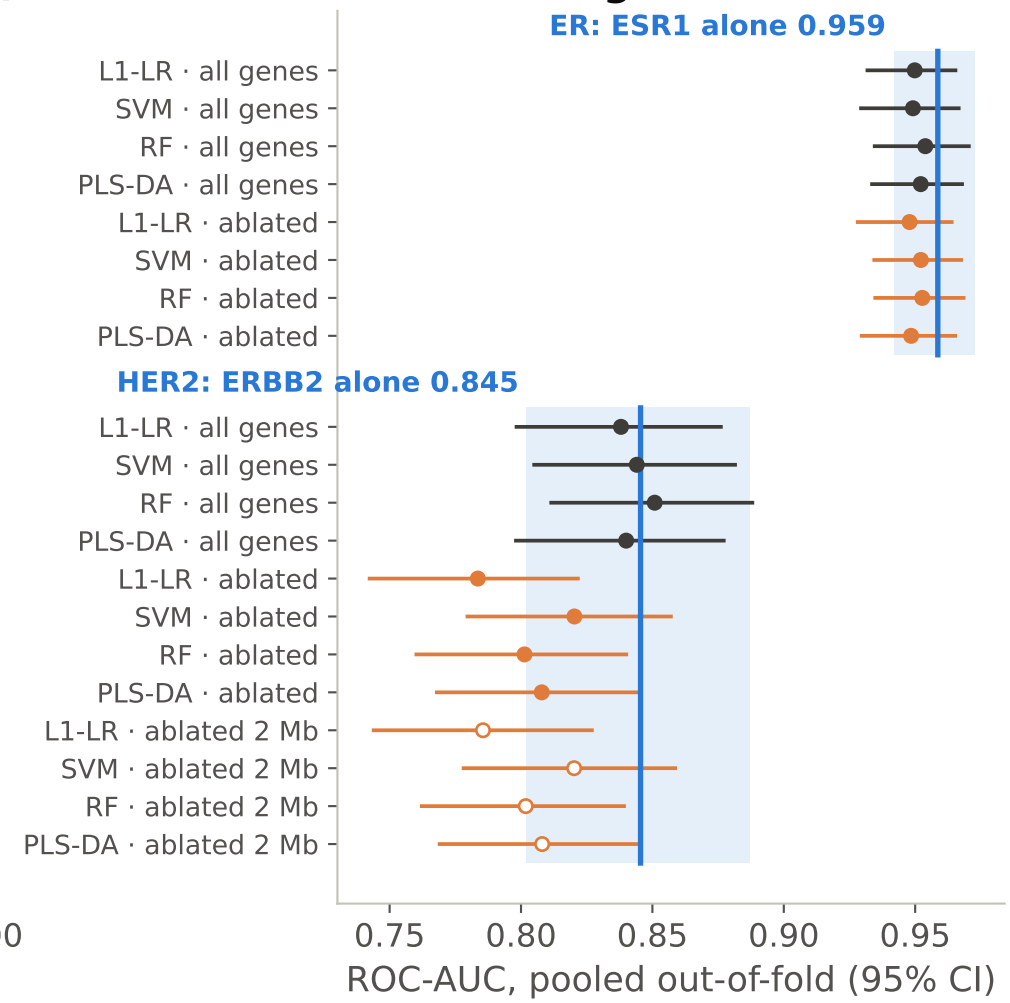
Clinical ER (IHC)



Clinical HER2 (IHC/FISH)



4 models × gene sets



ER 789+/235−, HER2 161+/752−. Nested CV: outer 5-fold × 3, inner 5-fold. Ablated: minus ESR1/PGR/ERBB2, genes ±1.5 Mb (or 2 Mb) of ERBB2, and PAM50.